

Figure 5: **Performance trends of test accuracy across prompts during prompt optimization with TextGRAD.**
Highlighted text shows unique instructions used to evaluate the novelty of ideas. The final test accuracy was 0.78125, so none of the optimized prompts (1 – 12) out-performed the original.

Prompt Number	Validation Accuracy	Prompt Text
Starting Prompt	0.8	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel' Respond in two lines: Review: <This is the review you will generate after comparing idea with related papers.> Classification:<novel or not novel label according to your review/rationale>
1	0	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel'. A novel idea introduces a unique, groundbreaking concept or approach not previously covered in the literature. A not novel idea reiterates or slightly modifies existing research. Consider the historical context and frequency of similar studies when making your classification. Focus on identifying unique, groundbreaking elements that differentiate the idea from existing research. Do not rely solely on keywords or the mention of a controlled setting to determine novelty. If the classification is ambiguous, indicate uncertainty and suggest a human review. Respond with only the classification label: 'novel' or 'not novel'.
2	0.7	You are a classifier. Classify the research idea as 'novel' or 'not novel' based on the related papers. Provide a brief review and directly state the classification. Review: <This is the review you will generate after comparing the idea with related papers.> Classification: <novel or not novel label according to your review/rationale> Example: Review: The idea is unique as it combines adaptive interfaces with AI explanations, which is not covered in the provided papers. Classification: novel.
3	0	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel'. Respond in two lines: Review: Provide a detailed review comparing the idea with related papers. Include specific examples and details from the referenced papers to justify your classification. Highlight both similarities and differences between the proposed idea and existing methodologies. Ensure your review is concise and precise, focusing on the main arguments. Classification: Based on the review, classify the idea as 'novel' or 'not novel' according to the following criteria: - Uniqueness of the approach - Originality of the application - Novelty of the results Provide specific references or evidence from the papers mentioned to support your classification. Use assertive language to clearly convey your classification.

Figure 6: contd. TextGrad Prompt Optimisation.

Prompt Number	Validation Accuracy	Prompt Text
4	0.7	You are a classifier. Classify the research idea as 'novel' or 'not novel' based on the related papers. Respond concisely: Review: <brief review> Classification: <novel or not novel>
5	0	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel'. Respond in two lines: **Review**: Provide a detailed review comparing the idea with related papers. Include specific examples and details from the referenced papers to justify your classification. Highlight both similarities and differences between the proposed idea and existing methodologies. Ensure your review is concise and precise, focusing on the main arguments. **Classification**: Based on the review, classify the idea as 'novel' or 'not novel' according to the following criteria: - Uniqueness of the approach - Originality of the application - Novelty of the results - Provide specific references or evidence from the papers mentioned to support your classification. - Use assertive language to clearly convey your classification.
7	0.3	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel' based on the following criteria: **Definition of Novelty** : - A 'novel' idea introduces a unique, groundbreaking concept, methodology, or significant improvement over existing work. - A 'not novel' idea closely aligns with existing work without significant innovation. **Contextual Instructions** : - If the idea involves common methodologies or well-known techniques, explicitly mention these aspects in your review. - Consider the historical context and frequency of similar studies when making your classification. **Comparative Analysis** : - Compare the proposed idea with existing systems or technologies mentioned in the related papers. Highlight similarities to justify the classification. **Evidence and Examples** : - Provide specific examples or evidence from the related papers that demonstrate the lack of novelty. **Structured Format** : - Use a structured format with sections such as 'Introduction,' 'Evaluation Criteria,' 'Comparative Analysis,' and 'Conclusion' to present your review. Respond in two lines: Review: <This is the review you will generate after comparing the idea with related papers.> Classification: <novel or not novel label according to your review/rationale>
8	0	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel'. Respond in two lines: - Review: Provide a detailed review that includes specific references to the related papers, highlighting similarities and differences. Include direct quotes or specific sections from the related papers that support your classification. - Classification: Use precise terminology to classify the idea as 'novel' or 'not novel' based on your review/rationale. Avoid vague terms and be specific in your justification.

Figure 7: contd. TextGrad Prompt Optimisation.

Prompt Number	Validation Accuracy	Prompt Text
9	0.6	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel'. Definition of Novel: An idea is 'novel' if it introduces a new concept, methodology, or significant improvement that is not already well-documented in the provided papers. Criteria for Evaluation: 1. Uniqueness of the approach. 2. Combination of elements. 3. Presence of similar frameworks in the literature. Instructions: 1. Compare the proposed idea's methodology, scope, and application with those described in the related papers. 2. Highlight specific aspects of the idea and compare them with the related papers. 3. Provide a detailed review based on the comparison. 4. Conclude with a classification of 'novel' or 'not novel' based on this comparison. Respond in the following format: Review: <This is the review you will generate after comparing the idea with related papers.> Classification: <novel or not novel label according to your review/rationale>
10	0.6	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel'. A novel idea introduces a unique, groundbreaking concept or approach not previously covered in the literature. A not novel idea reiterates or slightly modifies existing research. Consider the historical context and frequency of similar studies when making your classification. Respond in two lines: Review: <one-sentence review> Classification: <novel or not novel> Ensure your response is concise and uses simple language. Avoid unnecessary details.
11	0.6	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel'. Respond with: 1. Review: Provide a concise review in no more than two sentences, comparing the idea with related papers. Ensure your review includes a clear rationale for why the idea is classified as 'novel' or 'not novel'. Avoid using uncertain terms like 'appears' or 'seems'. 2. Classification: Use the term 'novel' or 'not novel' consistently based on your review. Example: Review: The proposed idea of developing a human-centric explainable AI system is novel because it uniquely combines explainable AI techniques with iterative improvement through human feedback and predictive models. Classification: novel
12	0.5	You are a classifier. Given a research idea and related papers, classify the idea as 'novel' or 'not novel'. Respond in two lines: Review: <Provide a detailed review comparing the idea with related papers, including specific examples and reasons for your classification. Mention existing tools or research that cover similar capabilities.> Classification: <novel or not novel label according to your review/rationale. Use the term 'novel' consistently in both your review and classification. Ensure your response is detailed yet concise, avoiding unnecessary verbosity.>